Adjoints and the Spectral Theorem

Math 5250 — Week 5

Adjoints

The Adjoint of a Linear Map

Let V and W be finite-dimensional inner product spaces (over $\mathbb{R}$ or $\mathbb{C}$), and let $f : V \to W$ be linear.

Proposition. There is a unique linear map $f^* : W \to V$ such that

$$\langle f(v), w \rangle = \langle v, f^*(w) \rangle$$

for all $v \in V$ and $w \in W$. The map f^* is the *adjoint* of f.

See Chapter 10 of Roman, *Advanced Linear Algebra*, for more details.

Proof of Existence

Let $v_1, \ldots, v_n$ and $w_1, \ldots, w_m$ be orthonormal bases for V and W. If f^* exists, it must satisfy

$$\langle f(v_i), w_j \rangle = \langle v_i, f^*(w_j) \rangle.$$

So define

$$f^*(w_j) = \sum_{i=1}^{n} a_i v_i, \qquad a_i = \overline{\langle f(v_i), w_j \rangle},$$

and extend linearly to W.

Proof of Existence (continued)

Then

$$\begin{aligned}\langle v_k, f^*(w_j)\rangle &= \sum_{i=1}^{n} \overline{a_i}\langle v_k, v_i\rangle \\ &= \overline{a_k} \\ &= \langle f(v_k), w_j\rangle.\end{aligned}$$

So f^* has the desired property, and the computation shows it is unique. $\square$

The Matrix of the Adjoint

If E is an orthonormal basis, then

$$[f^*]_E = \overline{[f]_E}^\top.$$

Indeed,

$$\begin{aligned}
\langle f(u), v\rangle &= ([f]_E[u]_E)^\top \overline{[v]_E} \\
&= [u]_E^\top [f]_E^\top \overline{[v]_E} \\
&= [u]_E^\top \overline{\overline{[f]_E}^\top [v]_E} \\
&= \langle u, f^*(v)\rangle.
\end{aligned}$$

Properties of the Adjoint

a. $(f+g)^* = f^* + g^*$ and $(af)^* = \overline{a} f^*$;
b. $f^{**} = f$;
c. if $V = W$, then $(fg)^* = g^* f^*$;
d. if f is invertible, then $(f^{-1})^* = (f^*)^{-1}$.

Example. For (c), assume $V = W$. Then

$$\langle f(g(v)), w \rangle = \langle g(v), f^*(w) \rangle = \langle v, g^* f^*(w) \rangle,$$

so $(fg)^* = g^* f^*$.

Adjoints and Orthogonality

Theorem. (Roman, Theorem 10.3) Let $f : V \to W$ with V, W finite-dimensional inner product spaces. Then:

1. $\ker(f^*) = \mathrm{im}(f)^\perp$;
2. $\mathrm{im}(f^*) = \ker(f)^\perp$;
3. f is injective if and only if f^* is surjective;
4. f is surjective if and only if f^* is injective;
5. $\ker(f^*f) = \ker(f)$ and $\ker(ff^*) = \ker(f^*)$;
6. $\mathrm{im}(f^*f) = \mathrm{im}(f^*)$ and $\mathrm{im}(ff^*) = \mathrm{im}(f)$;
7. if ρ is projection onto $\mathrm{im}(\rho)$ along $\ker(\rho)$, then ρ^* is projection onto $\ker(\rho)^\perp$ along $\mathrm{im}(\rho)^\perp$.

Some Proofs

(1) $w \in \ker(f^*) \iff \langle v, f^*(w)\rangle = 0$ for all $v \iff \langle f(v), w\rangle = 0$ for all $v \iff w \in \mathrm{im}(f)^\perp$.

(For $\Leftarrow$ in the first step: an inner product is definite, so $\langle v, f^*(w)\rangle = 0$ for all v forces $f^*(w) = 0$.)

(3) If f is injective, then $\mathrm{im}(f^*)^\perp = \ker(f) = 0$, so $\mathrm{im}(f^*) = V$.

(5) Clearly $\ker(f) \subset \ker(f^*f)$. If $f^*f(v) = 0$, then

$$0 = \langle f^*f(v), v\rangle = \langle f(v), f(v)\rangle,$$

so $f(v) = 0$.

Normal Operators

Unitary Diagonalizability

Definition. Let V be an n-dimensional inner product space. A linear map $f : V \to V$ is *unitarily diagonalizable* (*orthogonally diagonalizable* over $\mathbb{R}$) if V has an orthonormal basis of eigenvectors for f.

Equivalently,

$$V = E(\lambda_1) \oplus \cdots \oplus E(\lambda_k)$$

with the λ_j distinct, $E(\lambda_j)$ the λ_j-eigenspace of f, and $E(\lambda_i) \perp E(\lambda_j)$ for $i \neq j$.

Normal Operators

Suppose f is unitarily diagonalizable, with orthonormal eigenbasis $E = \{u_1, \ldots, u_n\}$ and $f(u_j) = \lambda_j u_j$. Then $[f^*]_E$ is diagonal with entries $\overline{\lambda_j}$, so $[f]_E$ and $[f^*]_E$ commute.

Definition. A linear map f is *normal* if $ff^* = f^*f$.

Proposition. Over $\mathbb{C}$, f is normal if and only if it is unitarily diagonalizable.

Over $\mathbb{R}$ the characterization is more complicated; see Roman, Theorem 10.19.

Elementary Properties

- The inverse of a normal operator is normal.
- If f is normal, so is f^2:

$$(f^2)^* f^2 = f^* f^* ff = fff^* f^* = f^2 (f^2)^*,$$

  and hence so is any polynomial $P(f)$.

Warning. A sum or product of two normal operators need not be normal.

Properties of a Normal Operator (1)

Let f be normal.

(a) If $f(v) = 0$, then $f^*(v) = 0$, because

$$\langle f(v), f(v)\rangle = \langle v, f^*f(v)\rangle = \langle v, ff^*(v)\rangle = \langle f^*(v), f^*(v)\rangle.$$

(b) If $f^k(v) = 0$ for some $k \geq 1$, then $f(v) = 0$.

Let $g = f^*f$, which is *self-adjoint*: $g^* = g$. If $g^k(v) = 0$ with $k > 1$, then

$$\langle g^k(v), g^{k-2}(v)\rangle = \langle g^{k-1}(v), g^{k-1}(v)\rangle = 0,$$

so $g^{k-1}(v) = 0$. By induction, $g(v) = 0$.

Properties of a Normal Operator (1, continued)

Now if $f^k(v) = 0$, then

$$g^k(v) = (f^*)^k f^k(v) = 0,$$

so $g(v) = f^* f(v) = 0$, so $\langle f(v), f(v) \rangle = 0$, and $f(v) = 0$.

(c) Applying (b) to $f - \lambda$ (also normal): the minimal polynomial of f is *square free*.

Properties of a Normal Operator (2)

(d) If $f(v) = \lambda v$, then $f^*(v) = \overline{\lambda} v$. (Apply (a) to the normal map $f - \lambda$.)

(e) Eigenvectors v, w with distinct eigenvalues λ, μ are orthogonal:

$$\lambda\langle v, w\rangle = \langle f(v), w\rangle = \langle v, f^*(w)\rangle = \langle v, \overline{\mu} w\rangle = \mu\langle v, w\rangle,$$

so $\langle v, w\rangle = 0$ when $\lambda \neq \mu$.

Normal $\iff$ Unitarily Diagonalizable over $\mathbb{C}$

Proof. If f is unitarily diagonalizable, then in an orthonormal eigenbasis $[f^*] = \overline{[f]}$ is diagonal, so f and f^* commute.

Conversely, if f is normal, its minimal polynomial is square free and splits into linear factors over $\mathbb{C}$, so f is diagonalizable.

Choose an orthonormal basis of each eigenspace. Distinct eigenspaces are orthogonal, so these combine to an orthonormal eigenbasis for V. $\square$

Normal Operators over $\mathbb{R}$

Remark. Over $\mathbb{R}$, the minimal polynomial of a normal operator is still square free, but may have irreducible quadratic factors.

So V decomposes into eigenspaces (for real eigenvalues) and subspaces where the minimal polynomial has degree 2.

Analyzing the degree-two pieces gives the full result; see Roman, Theorem 10.14.

Self-Adjoint Operators

Self-Adjoint Operators

Definition. A linear map f is *self-adjoint* if $f^* = f$. Self-adjoint maps are normal.

Over $\mathbb{R}$, self-adjoint operators are the more natural class.

Self-adjoint operators are closed under addition, multiplication by *real* scalars, inverses, and *real* polynomials.

If $E = \{u_i\}$ is orthonormal, f is self-adjoint iff $[f]_E$ is *Hermitian* ($[f]^\top = \overline{[f]}$; symmetric over $\mathbb{R}$):

$$[f]_{ij} = \langle f(u_i), u_j \rangle = \langle u_i, f(u_j) \rangle = \overline{\langle f(u_j), u_i \rangle} = \overline{[f]_{ji}}.$$

Real Eigenvalues

Proposition. If f is self-adjoint, all of its eigenvalues are real.

Proof. If $f(u) = \lambda u$, then

$$\lambda\langle u, u\rangle = \langle f(u), u\rangle = \langle u, f(u)\rangle = \overline{\lambda}\langle u, u\rangle,$$

so $\lambda = \overline{\lambda}$.

Over $\mathbb{R}$, u might not lie in V. Instead: in an orthonormal basis, $X = [f]_E$ is real symmetric, hence Hermitian, so its eigenvalues are real. The characteristic polynomial depends only on X, so all roots of the characteristic polynomial of f are real. $\square$

The Spectral Theorem

Combining: the minimal polynomial of a self-adjoint f is square free and splits into real linear factors (over $\mathbb{R}$ or $\mathbb{C}$). So f is diagonalizable, distinct eigenspaces are orthogonal, and each eigenspace has an orthonormal basis.

Proposition. (Spectral Theorem)

- Over $\mathbb{R}$, f is self-adjoint if and only if it is orthogonally diagonalizable.
- Over $\mathbb{C}$, f is self-adjoint if and only if it is unitarily diagonalizable with real eigenvalues.

Unitary Operators

Unitary and Orthogonal Operators

Definition. $f : V \to V$ is *unitary* (*orthogonal* over $\mathbb{R}$) if f is invertible and $f^* = f^{-1}$.

Then

$$\langle f(v), f(w)\rangle = \langle v, f^* f(w)\rangle = \langle v, w\rangle,$$

so f preserves the inner product and the norm $\|v\|^2 = \langle v, v\rangle$.

Orthonormal Bases

Proposition. If f is unitary and $u_1, \ldots, u_n$ is an orthonormal basis, then so is $f(u_1), \ldots, f(u_n)$. Conversely, if f carries one orthonormal basis to another, it is unitary.

Proof of the converse. Write $v = \sum a_i u_i$ and $w = \sum b_i u_i$. Then

$$\langle f(v), f(w) \rangle = \Big\langle \sum a_i f(u_i), \sum b_j f(u_j) \Big\rangle = \sum_{i,j} a_i \overline{b_j} \delta_{ij} = \langle v, w \rangle,$$

so f preserves inner products, hence is unitary (next slide). $\square$

Isometries

Proposition. If f is an invertible isometry,
i.e. $\langle f(v), f(w)\rangle = \langle v, w\rangle$ for all v, w, then f is unitary.

Proof. $\langle v, f^*f(w) - w\rangle = 0$ for all v, so $f^*f(w) = w$ for all w.
Hence $f^* = f^{-1}$. $\square$

Proposition. Over $\mathbb{R}$, f is orthogonal if and only if $\|f(v)\| = \|v\|$
for all v, by the *polarization identity*

$$\langle v, w\rangle = \frac{1}{4}(\|v + w\|^2 - \|v - w\|^2).$$

Eigenvalues of Unitary Operators

Proposition. The eigenvalues of a unitary operator have absolute value 1.

Proof. If $f(v) = \lambda v$ with $v \neq 0$, then

$$\langle v, v \rangle = \langle f(v), f(v) \rangle = \lambda \overline{\lambda} \langle v, v \rangle,$$

so $|\lambda|^2 = 1$. $\square$

Unitary Matrices

Proposition. A matrix A is unitary if and only if $\overline{A}^\top A = I$ (over $\mathbb{R}$: $A^\top A = I$). For unitary A:

- the rows (and columns) of A form an orthonormal set;
- the eigenvalues of A have absolute value 1;
- $|\det A| = 1$; over $\mathbb{R}$, $\det A = \pm 1$.

Projections and Spectral Resolution

Projections

A linear map $L : V \to V$ is a *projection* if $L^2 = L$. Then:

- L is the identity on $\mathrm{im}(L)$;
- $v - L(v) \in \ker(L)$, since $L(v - L(v)) = 0$, so

$$v = (v - L(v)) + L(v) \in \ker(L) + \mathrm{im}(L);$$

- if $v = L(x) \in \ker(L) \cap \mathrm{im}(L)$, then $v = L(x) = L^2(x) = L(v) = 0$.

So $V = \ker(L) \oplus \mathrm{im}(L)$.

Orthogonal Projections

Definition. A projection L is *orthogonal* if it is self-adjoint.

Proposition. A projection L is orthogonal if and only if $\ker(L) \perp \mathrm{im}(L)$.

Proof. ($\Rightarrow$) If $x = L(u) \in \mathrm{im}(L)$ and $y \in \ker(L)$:

$$\langle x, y \rangle = \langle L(u), y \rangle = \langle u, L(y) \rangle = 0.$$

Orthogonal Projections (continued)

($\Leftarrow$) Write $v = u + w$ and $v' = u' + w'$ with $u, u' \in \ker(L)$ and $w, w' \in \mathrm{im}(L)$. Then

$$\langle L(v), v' \rangle = \langle w, u' + w' \rangle = \langle w, w' \rangle = \langle u + w, w' \rangle = \langle v, L(v') \rangle.$$

So L is self-adjoint. $\square$

Spectral Resolution

If L is normal over $\mathbb{C}$ with orthogonal eigenspace decomposition

$$V = E(\lambda_1) \oplus \cdots \oplus E(\lambda_k),$$

let p_i be the orthogonal projection onto $E(\lambda_i)$. Then

$$L = \sum_{i=1}^{k} \lambda_i p_i$$

is the *spectral resolution* of L. A self-adjoint L over $\mathbb{R}$ has a similar resolution with real eigenvalues.

Proposition. Each p_i is a polynomial in L: if $f_i(x)$ satisfies $f_i(\lambda_i) = 1$ and $f_i(\lambda_j) = 0$ for $j \neq i$, then $p_i = f_i(L)$.

Positivity

Positive Operators

Definition. A self-adjoint linear map $L : V \to V$ is *positive* if

$$Q_L(v) = \langle L(v), v \rangle \geq 0$$

for all v, and *positive definite* if $Q_L(v) > 0$ for all $v \neq 0$.

Remark. $Q_L(v)$ is real, since

$$\overline{\langle L(v), v \rangle} = \langle v, L(v) \rangle = \langle L(v), v \rangle.$$

Proposition. L is positive if and only if all its eigenvalues are nonnegative, and positive definite if and only if they are all strictly positive.